Notes: Ito (AER, 2014)

Do Consumers Respond to Marginal or Average Price? Evidence from Nonlinear Electricity Pricing

Contents

1 Big picture **3**

2 Incremental contribution of the paper **3**

3 Data and measurement **4**

3.1 Theoretical price measures . . . 4

3.2 Research design and service-area border . . . 4

3.3 Billing data and sample . . . 5

3.4 Electricity tariffs and price variation . . . 5

4 Methodology and empirical specifications **6**

4.1 Bunching at kink points . . . 6

4.2 Encompassing test: marginal price versus average price . . . 6

4.3 Simulated instruments and controls . . . 6

4.4 Expected marginal price versus average price . . . 7

4.5 Direct estimation of the perceived price shape . . . 7

4.6 Welfare counterfactuals . . . 7

5 Identification and interpretation **8**

5.1 What the paper can identify . . . 8

5.2 Why the border design matters . . . 8

5.3 Price endogeneity and simulated instruments . . . 8

5.4 Parallel trends and sorting . . . 8

5.5 Interpretation of average-price behavior . . . 9

5.6 Limitations . . . 9

6 Tables and figures **9**

6.1 Figures 1 and 2: Theory and research-design geography . . . 9

6.2 Figures 3 and 4: Cross-sectional and time-series tariff variation . . . 10

6.3 Table 1 and Figure 5: Balance and bunching . . . 11

6.4 Figure 6 and Table 2: Difference-in-differences and the marginal-price test . . . 12

6.5 Tables 3 and 4: Lagged responses and expected marginal price . . . 13

6.6 Figure 7 and Table 5: Directly estimating the perceived price 14
6.7 Table 6 and Figure 8: Welfare implications 15

7 Short summary **16**

8 Conclusion **17**
8.1 Core conclusion 17
8.2 Economic intuition 17
8.3 Incremental contribution revisited 17
8.4 Implications 17
8.5 Limitations 17
8.6 Takeaway 17

1 Big picture

- **Research question.** When consumers face nonlinear electricity tariffs, do they respond to the marginal price, the expected marginal price, or the average price on the bill?
- **Main idea.** Standard theory says that consumers should optimize using marginal price. With uncertainty about final monthly usage, rational consumers should use expected marginal price. If nonlinear schedules are cognitively costly, consumers may instead use average price as a simple approximation.
- **Empirical setting.** The paper studies residential electricity customers near the border between Southern California Edison (SCE) and San Diego Gas & Electric (SDG&E) in Orange County, California. Households in the same cities face different nonlinear tariffs because the utility service boundary cuts through city limits.
- **Data.** The main data are household-level monthly billing records from 1999 to 2007 for 40,729 households that remain at the same premise throughout the sample period. The billing records are matched to Census block-group demographics using nine-digit zip codes.
- **Main finding.** Consumers respond strongly to average price, not to marginal price or expected marginal price. Once average price is controlled for, marginal-price and expected-marginal-price effects are statistically indistinguishable from zero.
- **Three empirical strategies.** The paper uses bunching at tariff kinks, encompassing tests of alternative price measures, and direct estimation of the perceived price schedule.
- **Core mechanism.** Nonlinear electricity bills are complex and consumers usually do not observe real-time cumulative usage. Average price requires only the total bill and total consumption, so it is much cheaper to understand.
- **Policy implication.** Increasing-block pricing is often justified as a conservation tool because high-use customers face high marginal prices. If consumers respond to average price instead, the conservation incentive is much weaker and can even slightly raise aggregate consumption relative to a revenue-neutral flat rate.
- **Economic takeaway.** The welfare and policy effects of nonlinear pricing depend critically on the price consumers actually perceive, not merely on the price that theory assumes they should use.

2 Incremental contribution of the paper

- **Field evidence on perceived prices.** Prior surveys and lab experiments suggested that people may confuse average and marginal prices. Ito provides field evidence using administrative billing data and actual utility tariffs.
- **A framework for perceived nonlinear prices.** The paper does not only test one behavioral story. It nests marginal price, expected marginal price, and average price in a general perceived-price framework.
- **Spatial discontinuity design.** The paper exploits a rare utility boundary that lies within city limits, allowing households in the same local housing and labor markets to face different nonlinear price schedules.

- **Substantial price variation.** SCE and SDG&E independently changed their tariffs over time, especially during and after the 2000–2001 California electricity crisis. This creates cross-sectional and time-series variation in marginal and average prices.
- **Separates marginal and average price effects.** The key empirical advantage is that the difference in marginal prices between SCE and SDG&E often moves differently from the difference in average prices. This makes it possible to distinguish competing price perceptions.
- **Household-level panel evidence.** The paper uses monthly household observations, not aggregate utility-level data, so it can construct actual marginal, expected marginal, and average prices for each billing period.
- **Welfare consequences.** The paper shows that behavioral price perception changes both the conservation effect and the deadweight-loss calculation for nonlinear pricing.
- **General relevance.** Although the empirical setting is electricity, the logic applies to income taxes, medical insurance, water pricing, phone plans, and other nonlinear price or tax schedules.

3 Data and measurement

3.1 Theoretical price measures

The paper considers three candidate prices that consumers may use when facing a nonlinear schedule $p(x)$.

- **Marginal price.** The standard nonlinear-budget-set model predicts that consumers respond to the price of the next unit at their chosen consumption level.
- **Expected marginal price.** If consumers are uncertain about final monthly consumption, a rational consumer should respond to the probability-weighted average of marginal prices across possible end-of-month consumption levels.
- **Average price.** If understanding the nonlinear schedule is costly, consumers may use total payment divided by total quantity as a simple approximation.

The paper writes a general perceived price as

$$\widetilde{p}(x) = \int p(x+e)w(e)\,de, \qquad \int w(e)\,de = 1.$$

Different weighting functions $w(e)$ generate the different cases. A point mass at zero gives marginal price. A distribution around the expected usage shock gives expected marginal price. A uniform weight over the units consumed gives average price.

3.2 Research design and service-area border

SCE serves a large part of Southern California, while SDG&E serves most of San Diego County and part of Orange County. In the study area, the SCE–SDG&E border lies within six Orange County cities: Laguna Beach, Laguna Niguel, Aliso Viejo, Laguna Hills, Mission Viejo, and Coto de Caza.

- Households generally cannot choose their electricity provider; the provider is determined by the address.
- The border was established before the relevant city boundaries, so it does not simply coincide with city-level policy or demographic differences.
- The same natural gas provider serves households on both sides, reducing concerns about differential substitution between gas and electricity.
- The analysis uses household panel variation, so time-invariant household differences are differenced out.

Interpretation. The border creates a quasi-experimental comparison: very similar households in the same local area experience different nonlinear price schedules.

3.3 Billing data and sample

The author obtains administrative billing records from SCE and SDG&E for 1999–2007. Each monthly record includes account and premise identifiers, billing start and end dates, monthly consumption, the bill amount, tariff type, climate zone, and nine-digit zip code.

- The main sample includes households on the default standard residential tariff.
- The main balanced panel contains 40,729 households that stay at the same premise throughout the sample.
- The Census match uses the nine-digit zip code to assign neighborhood characteristics at the block-group level.
- Table 1 shows that census characteristics and pre-treatment log electricity consumption in 1999 are closely balanced across the SCE and SDG&E sides of the border.

Interpretation. The sample is especially useful because it combines high-frequency billing data, precise tariff information, and local demographic balance.

3.4 Electricity tariffs and price variation

Residential electricity tariffs are increasing-block schedules. Monthly consumption is measured relative to a baseline amount of about 10 kWh per day. The marginal price changes discretely as consumption moves into higher tiers.

The key sources of variation are:

- **Cross-sectional variation.** In a given year, SCE and SDG&E customers can face different tariff schedules even within the same city.
- **Time-series variation.** The utilities changed their tariffs independently. SDG&E prices rose sharply in the summer of 2000 during the California electricity crisis, while SCE prices were initially protected. In 2001, SCE introduced a five-tier schedule and SDG&E followed later.
- **Different movements in marginal and average prices.** For some usage levels and years, SCE can have a higher marginal price while SDG&E has a higher average price, or vice versa. This divergence is central to identification.

4 Methodology and empirical specifications

4.1 Bunching at kink points

The first test asks whether consumers bunch at the kink points of the nonlinear price schedule. If consumers optimize on the true marginal price and have a nonzero elasticity, there should be excess mass around kinks where the marginal price jumps.

The paper compares SCE consumption distributions in 1999 and 2007. In 1999, the schedule is close to flat. In 2007, SCE has a steep five-tier schedule, including a marginal-price jump of more than 80% at one kink.

- The consumption distribution remains smooth in 2007.
- There is no visible bunching at any kink point.
- Applying existing bunching estimators gives marginal-price elasticities near zero.

Interpretation. The absence of bunching is inconsistent with consumers responding strongly to marginal price. It leaves open two possibilities: true marginal-price elasticity is near zero, or consumers respond to a smoother price such as average price.

4.2 Encompassing test: marginal price versus average price

The second strategy estimates whether consumption responds to marginal price, average price, or both. Let x_{it} be household i's average daily electricity use in billing month t. The core regression is

$$\Delta \ln x_{it} = \beta_1 \Delta \ln MP_{it} + \beta_2 \Delta \ln AP_{it} + f_t(x_{i,t_m}) + \gamma_{ct} + \delta_{bt} + u_{it},$$

where MP is marginal price, AP is average price, $t_m = t - 6$ is the middle period used to construct the simulated instrument, γ_{ct} are city-by-time fixed effects, and δ_{bt} are billing-cycle-by-time fixed effects.

The empirical logic is an encompassing test:

- If consumers respond to marginal price, average price should have no independent effect once marginal price is included.
- If consumers respond to average price, marginal price should have no independent effect once average price is included.

The key finding is that average price remains negative and significant, while marginal price becomes insignificant after controlling for average price.

4.3 Simulated instruments and controls

A major problem is that marginal and average prices are functions of consumption. The paper uses policy-induced price changes as instruments. For example, the instrument for a price change evaluates the new and old tariff schedules at a predetermined consumption level rather than at contemporaneous consumption.

The paper uses the middle-period consumption x_{i,t_m} to reduce mean-reversion concerns and adds very flexible controls for the consumption distribution:

- percentile-by-time fixed effects based on middle-period consumption;
- city-by-time fixed effects;
- billing-cycle-by-time fixed effects.

Interpretation. These controls allow high- and low-use households to have different time patterns, while the border comparison supplies the identifying price variation.

4.4 Expected marginal price versus average price

The paper also tests whether consumers respond to expected marginal price. The expected marginal price is calculated using an estimated uncertainty distribution for monthly consumption. A typical consumer can forecast monthly consumption with a standard error of about 20%.

The result is the same as in the marginal-price test. Expected marginal price is significant by itself, but it becomes insignificant when average price is included. Average price remains significant.

4.5 Direct estimation of the perceived price shape

Instead of assuming a specific perceived price, the paper estimates how consumers weight marginal prices around their actual consumption. Define

$$x_{it}^k = \left(1 + \frac{k}{100}\right) x_{it}, \qquad p_{it}^k = p(x_{it}^k), \qquad -100 \le k \le 100.$$

Consumers may use a weighted average of the surrounding marginal prices. The paper parameterizes weights $w_k(\alpha, \theta)$ and estimates

$$\Delta \ln x_{it} = \beta \sum_{k=-100}^{100} w_k(\alpha, \theta) \Delta \ln p_{it}^k + f_t(x_{i,t_m}) + \gamma_{ct} + \delta_{bt} + u_{it}.$$

The parameter α measures how much weight consumers put on prices below actual consumption versus above it. If consumers use expected marginal price, α should be around 0.5. If consumers use average price, α should be close to 1, since average price is determined by the prices of units already consumed.

The estimates of α are about 0.88–0.91. The paper rejects $\alpha = 0.5$ but cannot reject $\alpha = 1$. The estimated weighting function is close to the average-price weighting function.

4.6 Welfare counterfactuals

The welfare section compares nonlinear pricing with a revenue-neutral flat marginal rate. The paper asks what happens under two behavioral assumptions:

- consumers respond to average price, as estimated in the paper;
- consumers respond to marginal price, as standard theory assumes.

For SCE in 2007, nonlinear pricing slightly increases aggregate consumption by 0.27% relative to a counterfactual flat rate if consumers respond to average price. If consumers respond to marginal price, the same nonlinear schedule would reduce consumption by 2.33%.

The paper also computes deadweight loss under different assumptions about the social marginal cost of electricity. Average-price response can reduce or increase the deadweight loss relative to marginal-price response, depending on how high the social marginal cost is.

5 Identification and interpretation

5.1 What the paper can identify

The paper identifies the price measure that best explains household electricity consumption responses to nonlinear tariffs. The evidence supports average price as the perceived price. This is not only a statement about one coefficient; it is supported by three complementary pieces of evidence: no bunching at kinks, encompassing tests, and direct estimation of the perceived-price shape.

5.2 Why the border design matters

A usual nonlinear-pricing study has a core identification problem: all similar consumers often face the same schedule, so the researcher cannot separate price effects from distributional changes in consumption. Here, households in the same cities and climate zone face different tariff schedules because of the service-area border.

The border design helps because:

- local economic and weather shocks can be absorbed by city-by-time controls;
- household fixed differences are removed by comparing each household to its own same-month prior-year consumption;
- relative tariff changes vary across utilities and over time;
- demographic and 1999 consumption balance reduces sorting concerns.

5.3 Price endogeneity and simulated instruments

Observed marginal price and average price are mechanically related to consumption. A household with a high demand shock may move to a higher tier, making the observed marginal price endogenous. The simulated-instrument approach addresses this by applying tariff changes to predetermined consumption.

Interpretation. The instrument captures how the price schedule changes for a given consumption level, not how contemporaneous demand shocks move households across the schedule.

5.4 Parallel trends and sorting

Figure 6 provides a visual check on the parallel trend assumption. Before the major relative price divergence, the SCE and SDG&E consumption trends for the relevant group are similar. Table 1

also shows that census variables and 1999 log consumption are balanced across the border.

Sorting based on expected electricity bills is unlikely for two reasons. First, customers cannot choose their utility except by moving. Second, the relative price advantage switches over time, so it would be difficult to choose a side based on future expected bills.

5.5 Interpretation of average-price behavior

Average-price response is not necessarily irrational in a narrow sense. Electricity bills are complex, consumers usually do not observe cumulative monthly usage in real time, and the marginal price depends on both the tariff schedule and the final consumption tier. Average price can be calculated from two salient pieces of information on the bill: total payment and total consumption.

The paper therefore interprets average-price behavior as a form of suboptimization or inattention to complex schedules, rather than as a failure to respond to prices altogether.

5.6 Limitations

The paper studies residential electricity in one region and one period. Consumers may respond differently when real-time feedback or smart-meter information is available. The welfare calculations rely on demand-elasticity estimates and assumptions about private and social marginal costs. Expected marginal price is constructed using an estimated uncertainty distribution, although the direct weighting-function test helps address this concern.

6 Tables and figures

6.1 Figures 1 and 2: Theory and research-design geography

Read Figure 1:

- The figure shows the three candidate perceived prices in a simple nonlinear schedule: marginal price, expected marginal price, and average price.
- Panel A illustrates that marginal price jumps sharply at a kink, while expected marginal and average price are smoother.
- Panel B shows the weighting functions that convert the true marginal price schedule into each perceived price.

Read Figure 2:

- The map shows the SCE–SDG&E border in Orange County.
- The border cuts through city limits in six cities, so households in the same city can face different utility tariffs.
- SCE serves the north side and SDG&E serves the south side of the border.

Economic message: The theory figure defines the competing behavioral models, while the map shows why the paper has unusually clean variation for testing them.

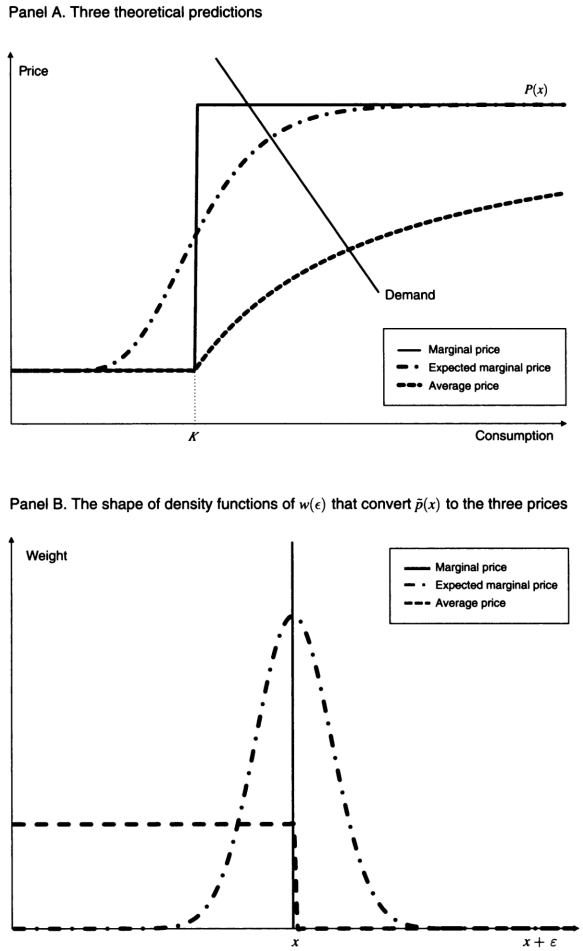

FIGURE 1. THEORETICAL PREDICTIONS ABOUT CONSUMERS' PERCEIVED PRICE

Notes: Panel A uses a simple example of nonlinear price schedules to describe the three theoretical predictions about perceived price. Panel B shows the density functions of $w(\epsilon)$ that convert $\tilde{p}(x)$ to the the corresponding prices presented in panel A.

Figure 1: Theoretical predictions about consumers' perceived price.

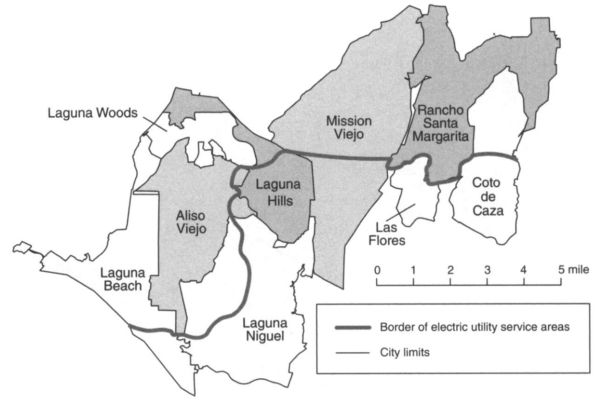

FIGURE 2. BORDER OF ELECTRICITY SERVICE AREAS IN ORANGE COUNTY, CALIFORNIA

Notes: The border of electricity service areas lies within city limits in six cities. SCE serves the north side of the border and SDG&E serves the south side of the border.

Figure 2: Border of electricity service areas in Orange County, California.

6.2 Figures 3 and 4: Cross-sectional and time-series tariff variation

Read Figure 3:

- The figure compares the 2002 tariff schedules for SCE and SDG&E.
- Solid lines show marginal price schedules; dashed lines show average prices.
- At some consumption levels, the utility with the higher marginal price is not the utility with the higher average price.

Read Figure 4:

- The figure shows how SCE and SDG&E residential prices changed over time.
- SDG&E prices rise sharply in 2000 during the electricity crisis.

- SCE introduces a five-tier schedule in 2001, and both utilities make further independent changes afterward.

Economic message: The separate movement of marginal and average prices is what lets the paper distinguish the price consumers actually respond to.

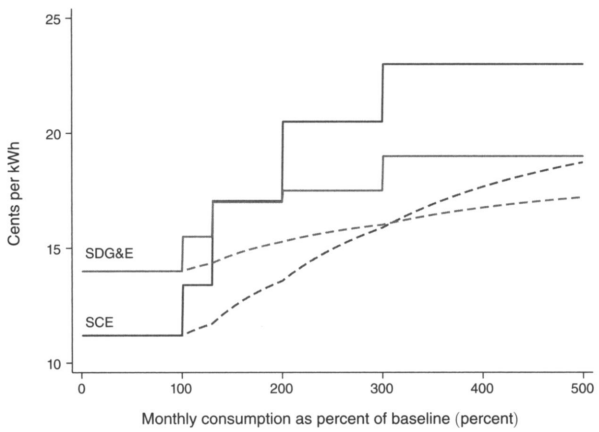

FIGURE 3. AN EXAMPLE OF CROSS-SECTIONAL PRICE VARIATION IN NONLINEAR ELECTRICITY PRICING

Note: To show an example of cross-sectional price variation, this figure presents the marginal price (solid line) and the average price (dashed line) for SCE and SDG&E in 2002.

Figure 3: Cross-sectional variation in nonlinear electricity pricing.

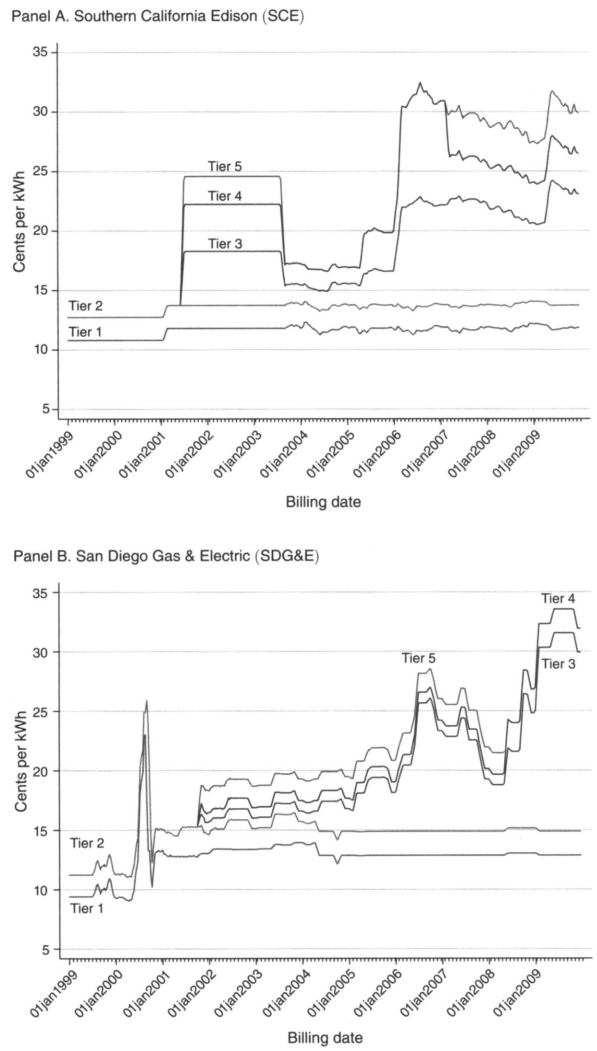

FIGURE 4. TIME-SERIES PRICE VARIATION IN NONLINEAR ELECTRICITY PRICING

Figure 4: Time-series variation in nonlinear electricity pricing.

6.3 Table 1 and Figure 5: Balance and bunching

Read Table 1:

- The table compares SCE and SDG&E households in the six border cities.
- Demographic and housing variables are broadly similar across the two sides.
- Log electricity use in 1999 is essentially identical across the two utility areas.

Read Figure 5:

- The figure compares SCE consumption distributions in 1999 and 2007.
- In 2007, the nonlinear schedule has sharp kink points and large marginal-price jumps.
- The consumption distribution remains smooth; there is no visible bunching around the kinks.

Economic message: The balance table supports the border comparison, and the bunching figure gives the first evidence against strong marginal-price optimization.

Table 1—Summary Statistics and Differences in Means

	SCE		SDG&E		Difference	
	Mean	(SE)	Mean	(SE)	Mean	(SE)
Data from census 2000						
Income per capita ($)	40,773	(1,591)	40,832	(1,627)	59	(2,261)
Median home value ($)	391,508	(19,987)	404,887	(19,768)	13,379	(27,849)
Median rent ($)	1,364	(41)	1,385	(62)	21	(74)
Population density/mile^2	6,084	(362)	5,423	(360)	−662	(508)
Household size	2.71	(0.07)	2.81	(0.05)	0.11	(0.09)
Median age	47.71	(1.23)	45.73	(0.55)	−1.98	(1.35)
Percent owner occupied housing	81.86	(1.65)	84.27	(1.93)	2.41	(2.53)
Percent male	49.12	(0.41)	48.65	(0.32)	−0.46	(0.52)
Percent employment of males	74.90	(2.14)	78.67	(1.13)	3.78	(2.41)
Percent employment of females	57.75	(1.83)	58.54	(1.22)	0.79	(2.19)
Percent college degree	50.31	(1.28)	52.96	(1.22)	2.65	(1.76)
Percent high school degree	35.25	(1.11)	32.27	(0.93)	−2.98	(1.44)
Percent no high school degree	4.28	(0.29)	4.07	(0.33)	−0.21	(0.44)
Percent white	85.53	(0.86)	83.74	(0.94)	−1.79	(1.27)
Percent Hispanics	9.33	(0.58)	9.70	(0.74)	0.37	(0.93)
Percent Asian	6.97	(0.61)	8.23	(0.66)	1.26	(0.90)
Percent black	1.19	(0.15)	0.86	(0.16)	−0.32	(0.22)
Electricity billing data						
Electricity use (kWh/day)	21.37	(0.07)	22.48	(0.09)	1.11	(0.12)
ln(electricity use)	2.89	(0.00)	2.89	(0.01)	0.00	(0.00)
ln(electricity use) in 1999	2.86	(0.00)	2.86	(0.01)	0.01	(0.01)

Notes: For each variable, I show the mean and standard error for SCE customers and SDG&E customers in the six cities that have the territory border of SCE and SDG&E within the city limits. The last column shows the difference in the mean with the standard error of the difference. I cluster standard errors at the census block group level for the census data and at the customer level for the electricity billing data.

Table 1: Summary statistics and differences in means.

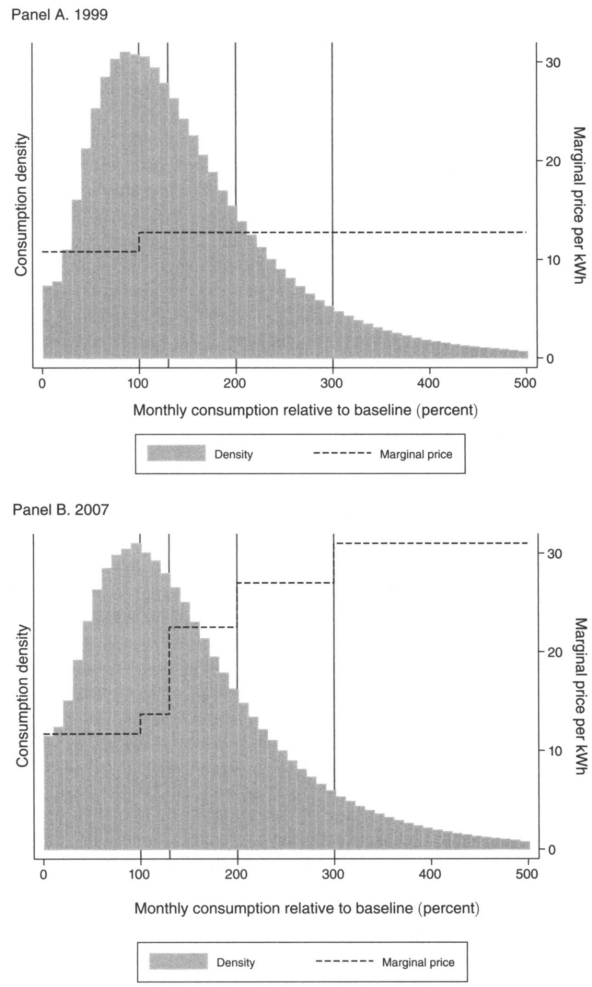

Figure 5. Consumption Distributions and Nonlinear Price Schedules

Notes: The figure shows the histogram of household-level monthly electricity consumption for SCE in 1999 (panel A) and in 2007 (panel B). The figure also shows the nonlinear price schedule for each year. The vertical solid lines show the kink points of the nonlinear price schedule.

Figure 5: Consumption distributions and nonlinear price schedules.

6.4 Figure 6 and Table 2: Difference-in-differences and the marginal-price test

Read Figure 6:

- The figure plots difference-in-differences in prices and consumption for January relative to 1999 among fourth-tier consumers.
- Predicted and actual price changes move together, supporting the first-stage logic of the instruments.
- In years where marginal and average price differences move in opposite directions, consumption follows average price more closely.

Read Table 2:

- Columns with only marginal price show a negative marginal-price coefficient.
- Once average price is included, the marginal-price coefficient becomes small and insignificant.
- Average price remains negative and economically meaningful, both contemporaneously and with a one-month lag.

Economic message: Marginal price appears to matter only when average price is omitted. The encompassing test points to average price as the relevant perceived price.

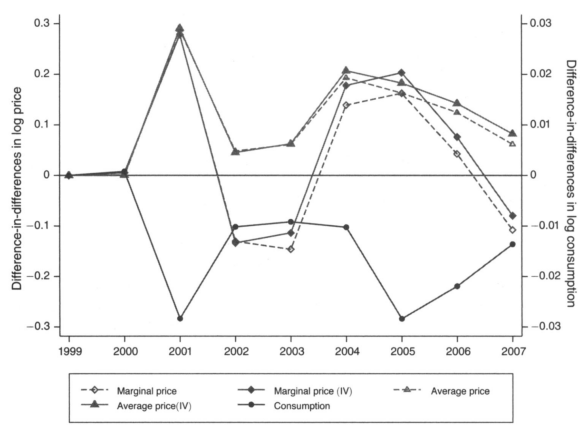

FIGURE 6. DIFFERENCE-IN-DIFFERENCES IN PRICE AND CONSUMPTION

Notes: The figure shows the difference-in-differences (DD) in price and consumption for the billing month of January relative to year 1999 for customers whose consumption lies in the fourth tier of the five-tier price schedule. For example, the plot of marginal price presents how SDG&E's log marginal price evolved from 1999 relative to that of SCE. First, for each side of the border, I calculate the mean log change in price and consumption. Then, I cal-

Figure 6: Difference-in-differences in price and consumption.

TABLE 2—ENCOMPASSING TESTS: MARGINAL PRICE VERSUS AVERAGE PRICE

	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta \ln(\textit{marginal price}_t)$	−0.034		0.002			
	(0.004)		(0.011)			
$\Delta \ln(\textit{average price}_t)$		−0.051	−0.054			
		(0.005)	(0.015)			
$\Delta \ln(\textit{marginal price}_{t-1})$				−0.050		0.006
				(0.004)		(0.011)
$\Delta \ln(\textit{average price}_{t-1})$					−0.074	−0.082
					(0.005)	(0.015)

Notes: This table shows the results of the IV regression in equation (3) with fixed effects and control variables specified in the equation. The unit of observation is household-level monthly electricity usage. The dependent variable is the log change in electricity consumption in billing period t from billing period $t-12$. The sample period is from January 1999 to December 2007 and the sample size is 3,752,378. Standard errors in parentheses are clustered at the household level to adjust for serial correlation.

Table 2: Encompassing tests - marginal price versus average price.

6.5 Tables 3 and 4: Lagged responses and expected marginal price

Read Table 3:

- Consumers respond more to lagged average prices than to contemporaneous average prices.
- The medium-long-run elasticity is around -0.071 to -0.088 as one- to four-month lagged average prices are included.
- The elasticity stabilizes after several months, suggesting that most of the delayed response is captured by short lags.

Read Table 4:

- Expected marginal price is significant when entered alone.
- Once average price is included, expected marginal price becomes statistically insignificant.
- Average price remains negative and significant.

Economic message: The results rule out the main rational-uncertainty alternative. The consumer response is better described by average price than by expected marginal price.

TABLE 3—LAGGED RESPONSES AND MEDIUM-LONG RUN PRICE ELASTICITY

		Medium-long run responses			
	Lagged responses (1)	1 month (2)	2 month (3)	3 month (4)	4 month (5)
$\Delta \ln(\textit{average price}_t)$	0.001 (0.002)				
$\Delta \ln(\textit{average price}_{t-1})$	−0.049 (0.006)				
$\Delta \ln(\textit{average price}_{t-2})$	−0.026 (0.007)				
$\Delta \ln(\textit{average price}_{t-3})$	−0.011 (0.006)				
$\Delta \ln(\textit{average of lag average prices})$		−0.071 (0.005)	−0.082 (0.005)	−0.087 (0.006)	−0.088 (0.006)

Notes: See notes in Table 3. The dependent variable is the log change in electricity consumption in billing period t from billing period $t-12$. Because the four-month lag price is unknown for the first four months of the sample period, I include monthly bills from May 1999 to December 2007. The sample size is 3,598,571. Standard errors in parentheses are clustered at the household level to adjust for serial correlation.

Table 3: Lagged responses and medium-long-run price elasticity.

TABLE 4—ENCOMPASSING TESTS: EXPECTED MARGINAL PRICE VERSUS AVERAGE PRICE

	(1)	(2)	(3)	(4)
$\Delta \ln(\textit{expected marginal price}_t)$	−0.036 (0.004)	0.004 (0.012)		
$\Delta \ln(\textit{average price}_t)$		−0.056 (0.015)		
$\Delta \ln(\textit{expected marginal price}_{t-1})$			−0.053 (0.004)	0.009 (0.012)
$\Delta \ln(\textit{average price}_{t-1})$				−0.086 (0.015)

Notes: See notes in Table 3. This table shows the results of the IV regression in equation (3) with expected marginal price instead of marginal price. The dependent variable is the log change in electricity consumption in billing period t from billing period $t-12$. The data include monthly bills from January 1999 to December 2007. The sample size is 3,752,378. Standard errors in parentheses are clustered at the household level to adjust for serial correlation.

Table 4: Encompassing tests - expected marginal price versus average price.

6.6 Figure 7 and Table 5: Directly estimating the perceived price

Read Figure 7:

- The dashed lines show example weighting functions for how consumers might combine surrounding marginal prices.
- The solid line is the estimated weighting function.
- The estimated function places most weight on prices for units up to actual consumption and little weight above actual consumption.

Read Table 5:

- The weighting parameter α is about 0.91, 0.90, and 0.88 across price specifications.
- The paper rejects $\alpha = 0.5$, which would be consistent with symmetric expected-marginal-price weighting.
- The paper cannot reject $\alpha = 1$, which is consistent with average-price weighting.
- The elasticity estimates are between about -0.059 and -0.094.

Economic message: The direct perceived-price estimation independently confirms the encompassing-test result: consumers behave as if they use average price.

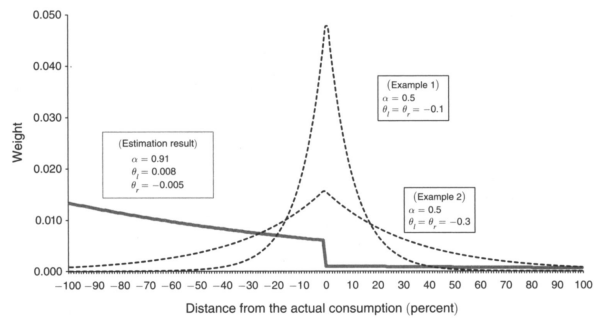

FIGURE 7. SHAPE OF WEIGHTING FUNCTIONS: EXAMPLES AND ESTIMATION RESULTS

Notes: The dashed lines illustrate two examples of the weighting functions of equation (5). The first example shows the case with $\alpha = 0.5$ and $\theta_l = \theta_r = -0.1$, in which consumers care about the price to the left and right of x_{it} equally but put larger weights on the price close to x_{it}. The second example shows a similar case but consumers care about the price further away from x_{it} The solid line shows the estimation result of $w(\alpha, \theta)$, which is shown in Table 5.

Figure 7: Shape of weighting functions - examples and estimation results.

TABLE 5—ESTIMATION OF THE SHAPE OF PERCEIVED PRICE

	Price variable		
	Current month (1)	One-month lag (2)	Four-month average (3)
Weighting parameter α	0.911	0.896	0.883
	(0.082)	(0.083)	(0.087)
Slope parameter θ_l	0.008	0.013	0.015
	(0.013)	(0.014)	(0.014)
Slope parameter θ_r	−0.005	−0.009	0.001
	(0.015)	(0.015)	(0.017)
Elasticity parameter β	−0.059	−0.086	−0.094
	(0.005)	(0.006)	(0.006)
p-value for H_0: $\alpha = 0.5$	0.00	0.00	0.00
p-value for H_0: $\alpha = 1$	0.28	0.21	0.18

Notes: See notes in Table 3. This table shows the results of the nonlinear IV regression in equation (5). Standard errors in parentheses are clustered at the household level to adjust for serial correlation. Column 1 uses the contemporaneous price; column 2, the one-month lagged price; and column 3, the average of one-, two-, three-, and four-month lagged prices, as a price variables in the regression.

Table 5: Estimation of the shape of perceived price.

6.7 Table 6 and Figure 8: Welfare implications

Read Table 6:

- If consumers respond to average price, nonlinear pricing raises aggregate consumption by 0.27% relative to a revenue-neutral flat rate.
- If consumers respond to marginal price, the same nonlinear tariff would reduce aggregate consumption by 2.33%.
- The conservation value of nonlinear pricing therefore depends on the perceived price, not just the statutory marginal price.

Read Figure 8:

- The figure compares deadweight loss when consumers respond to average price versus marginal price.
- When the social marginal cost is below about 21 cents/kWh, marginal-price response produces larger deadweight loss.
- When the social marginal cost is high, average-price response can produce larger deadweight loss because consumers do not reduce consumption enough.

Economic message: Behavioral price perception changes the sign and size of policy effects. Nonlinear pricing is not automatically an effective conservation policy when consumers perceive average prices.

Table 6—Effect of Nonlinear Pricing on Energy Conservation

	Assumption on consumers' perceived price	
	Average price	Marginal price
(A) Consumption under five-tier nonlinear pricing	20,526	19,993
(B) Consumption under counterfactual flat rate	20,471	20,471
Percent change from (B) to (A)	0.27 (0.02)	−2.33 (0.05)

Notes: The table shows how nonlinear pricing changes aggregate consumption compared to the counterfactual flat marginal rate for two scenarios: (i) customers respond to average price and (ii) customers respond to marginal price. This table uses data for SCE in 2007, where consumers had one of the steepest five-tier price schedules in the sample period. The results do not change when I use the data for other years or for SDG&E. Asymptotic standard errors are calculated by the delta method based on the standard errors of the estimated price elasticity.

Table 6: Effect of nonlinear pricing on energy conservation.

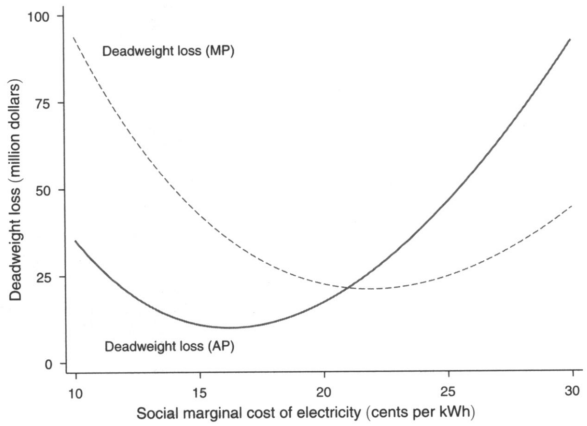

Figure 8. Efficiency Costs of Nonlinear Pricing

Notes: This figure presents the deadweight loss (DWL) from the five-tier tariffs for SCE in 2007, for different assumptions on the social marginal cost of electricity as well as on how consumers respond to nonlinear pricing. The DWL is calculated with the price elasticity of −0.088. The solid line shows the DWL when consumers respond to their average price. The dashed line displays a counterfactual DWL when consumers respond to their marginal price. The DWL is larger for the marginal price response when the social marginal cost is less than

Figure 8: Efficiency costs of nonlinear pricing.

7 Short summary

- The paper asks whether consumers facing nonlinear electricity pricing respond to marginal price, expected marginal price, or average price.
- The empirical setting is the SCE–SDG&E service-area border in Orange County, where households in the same cities face different nonlinear tariffs.
- The data are household-level monthly electricity billing records from 1999 to 2007, matched to Census demographics.
- The paper first shows no bunching at kinks, even when marginal prices jump sharply.
- Encompassing tests show that average price predicts consumption, while marginal price does not once average price is included.
- Expected marginal price also loses significance after controlling for average price.
- Direct estimation of the perceived-price weighting function gives a shape that is nearly identical to average price.
- Consumers appear to use average price because it is much easier to infer from the bill than the true marginal price.
- Nonlinear pricing does not necessarily conserve energy: under average-price response, the 2007 SCE nonlinear tariff slightly increases aggregate consumption relative to a revenue-neutral flat rate.
- The welfare cost of nonlinear pricing depends on social marginal cost and on the price measure consumers actually perceive.

8 Conclusion

8.1 Core conclusion

Ito shows that residential electricity consumers facing nonlinear tariffs respond to average price rather than marginal or expected marginal price. The result appears across bunching evidence, instrumental-variable encompassing tests, and direct perceived-price estimation.

8.2 Economic intuition

A nonlinear electricity tariff requires consumers to know their current cumulative usage, forecast end-of-month usage, identify the relevant tier, and map usage into a marginal price. In the sample period, most customers did not have real-time usage information. Average price is much easier: it can be calculated from total payment and total consumption on the bill. Consumers therefore act as if they simplify the nonlinear schedule into an average unit price.

8.3 Incremental contribution revisited

The paper's main contribution is to convert a long-standing behavioral question into a field test with clean price variation. It does not merely document that demand slopes downward. It identifies which price in a nonlinear schedule consumers appear to use.

8.4 Implications

The findings matter for policy design. Increasing-block tariffs, nonlinear taxes, subsidies, insurance contracts, and utility bills often rely on marginal incentives. If consumers perceive average prices, policy instruments that are designed to operate through marginal prices may have weaker or different effects than intended. Information provision, bill design, and real-time feedback may therefore be central parts of nonlinear pricing policy.

8.5 Limitations

The evidence comes from residential electricity consumers in Southern California during 1999–2007. Price perception may differ in settings where consumers have better real-time information or stronger incentives to optimize. The welfare calculations also require assumptions about elasticities and marginal costs. Still, the empirical evidence strongly rejects the maintained assumption that consumers automatically respond to marginal price.

8.6 Takeaway

The enduring takeaway is that nonlinear prices work through perceived incentives. In this setting, households perceive the average price on the bill, not the marginal price embedded in the tariff schedule. Policy makers should therefore design nonlinear pricing systems with consumer attention, information, and bill salience in mind.